

BIOS640-Week3-UNAssignment-Data-Visualization

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2026-09-24

Initial Setup

```
# Loading necessary packages
library(pacman)
pacman::p_load(tidyverse)
pacman::p_load(here)
pacman::p_load(rio)
pacman::p_load(knitr)
pacman::p_load(cowplot)
pacman::p_load(ggpubr)
pacman::p_load(dplyr)
pacman::p_load(DT)
pacman::p_load(RColorBrewer)

# Importing data
nhanes_data <- import(here("Data", "cleaned_NHANES.csv"))
diet_data <- import(here("Data", "diet.csv"))

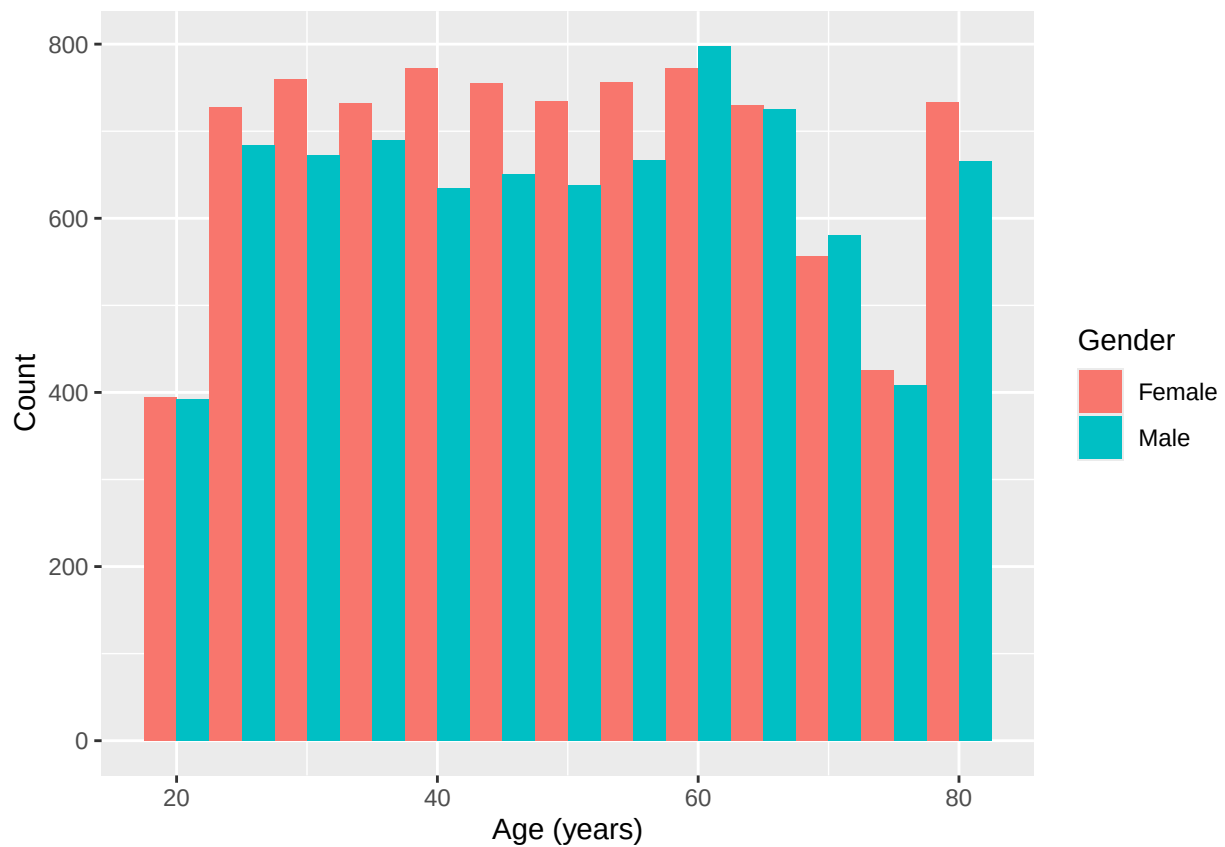
# Initial review of the data
dim(nhanes_data)
str(nhanes_data)
View(nhanes_data)
head(nhanes_data)
```

1. Reproducing and Arranging a Ggplot2 Figure

The gender distribution of patients across age groups and ethnicities was illustrated using two figures created with various ggplot2 tools. The figures were then combined using two different methods. The first method used the `plot_grid()` function from the `cowplot` package, while the second used the `ggarrange()` function from the `ggpubr` package which provides an easy way to create a shared legend. Although the overall layout is relatively simple, the combined figure with a shared legend provides a clearer and more cohesive presentation of the results.

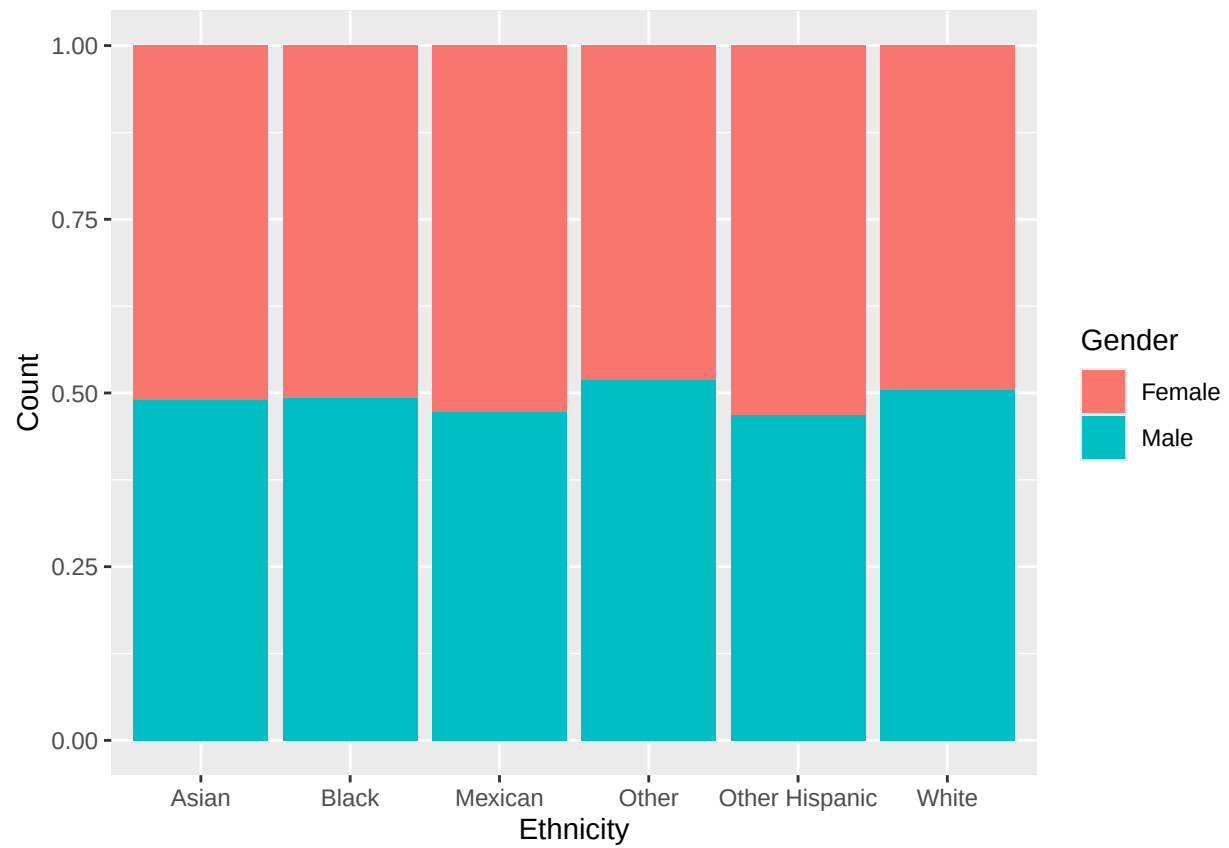
Individual Figures

```
# Gender distribution
p1 <- ggplot(
  nhanes_data %>% filter(age >= 20, age <= 80),
  aes(x = age, fill = gender)
) +
  geom_histogram(bins = 13, position = "dodge") +
  scale_x_continuous(breaks = seq(20, 80, by = 20)) +
  labs(
    x = "Age (years)",
    y = "Count",
    fill = "Gender"
  ) +
  theme_gray()
p1
```



```
# Ethnicity distribution
p2 <- ggplot(
  nhanes_data, aes(x = ethnicity_2, fill = gender)
) +
  geom_bar(position = "fill") +
  labs(
    x = "Ethnicity",
    y = "Count",
    fill = "Gender"
  ) +
  theme_gray()
```

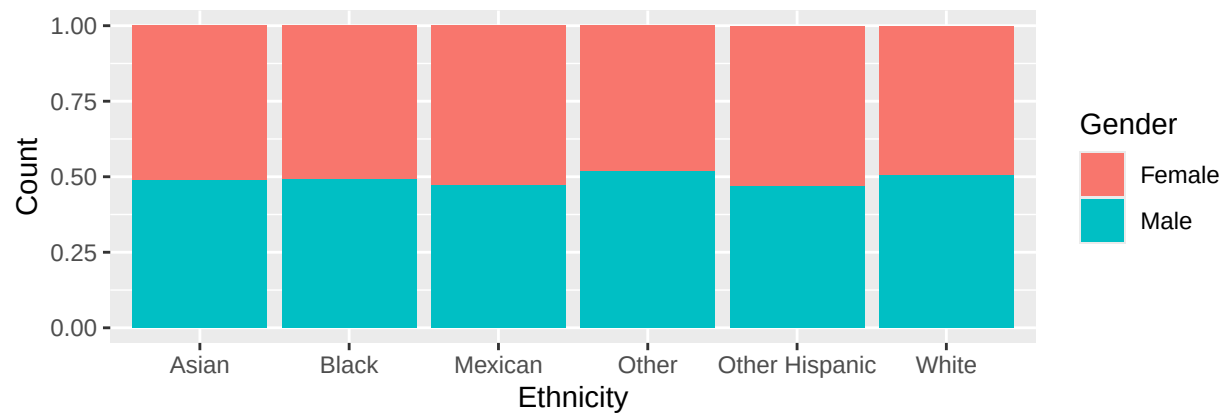
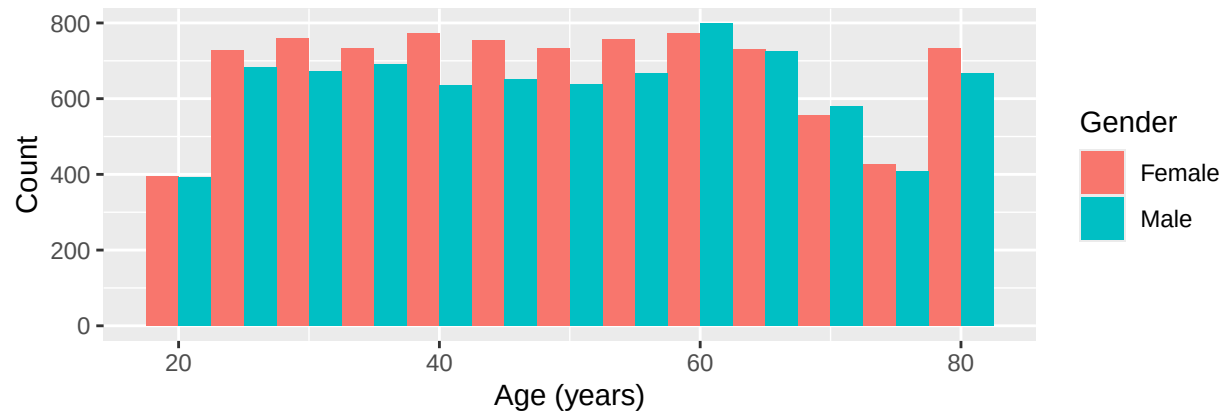
p2



Combined Figures

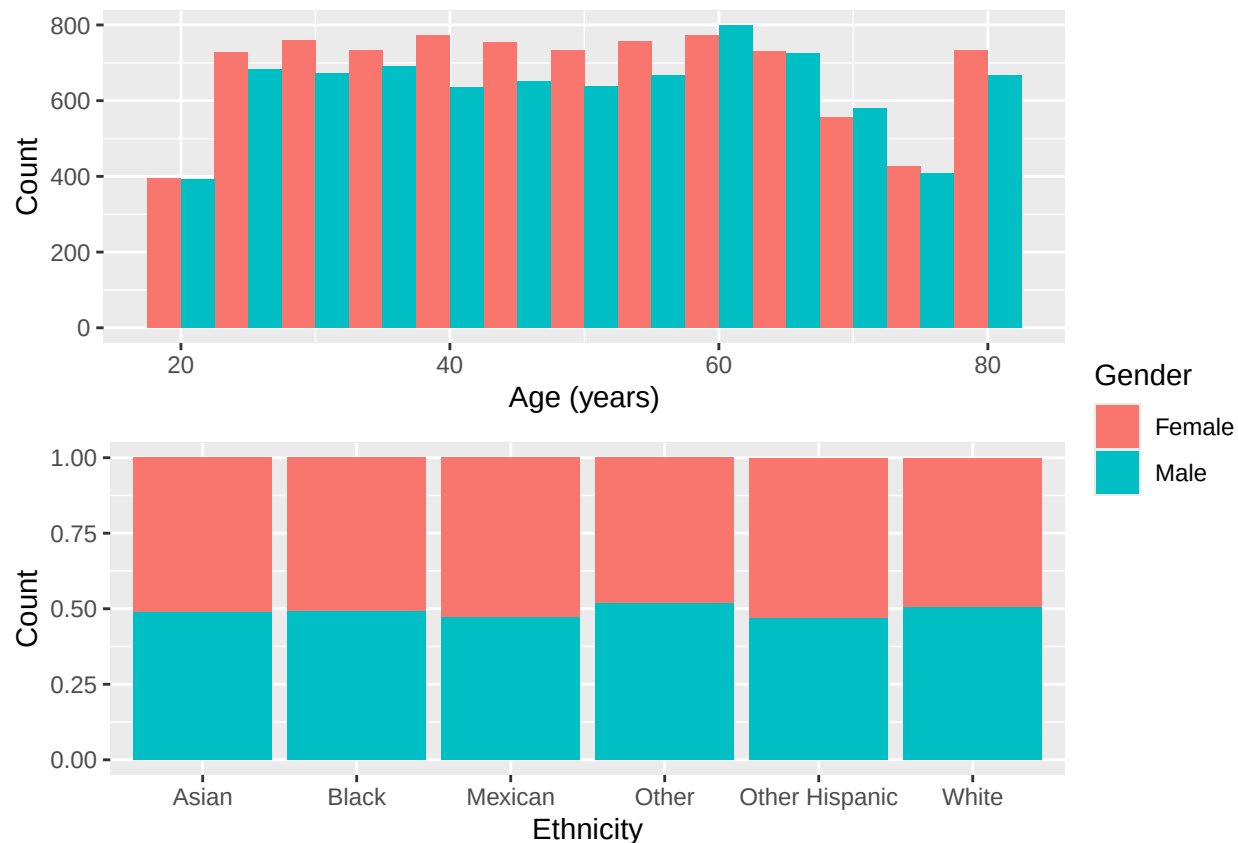
```
# Combining plots
# Using cowplot()
cowplot_combined_figure <- plot_grid(
  p1,
  p2,
  nrow = 2
)

cowplot_combined_figure
```



```
# Using ggpubr()
ggpubr_combined_figure <- ggarrange(p1, p2, nrow = 2,
  common.legend = TRUE, legend = "right")

ggpubr_combined_figure
```



2. Visualizing Key Demographic Characteristics

The absence of missing values for the age, gender and both ethnicity variables was confirmed using `summarise()`.

```
# Verifying the number of missing values
nhanes_data %>%
  summarise(
    age      = sum(is.na(age)),
    gender   = sum(is.na(gender)),
    ethnicity_1 = sum(is.na(ethnicity_1)),
    ethnicity_2 = sum(is.na(ethnicity_2))
  )
```

```
##   age gender ethnicity_1 ethnicity_2
## 1    0      0           0           0
```

The age distribution shows a higher number of participants under 20 years old, followed by a relatively even distribution among participants between 20 and 60 years old. The number of participants then decreases after age 60, with a last spike at 80 years old which can be explain by the fact that every patient older than that was entered in the 80 years old category. .

The gender distribution is relatively balanced, with slightly more female than male participants.

For both ethnicity variables, Non-Hispanic White participants represent the largest group, followed by non-Hispanic Black participants and Mexican American participants. In the first ethnicity variable (RIDRETH1), the remaining participants are primarily classified as Other Race, followed by Other Hispanic. The second ethnicity variable (RIDRETH3) includes an additional category for Non-Hispanic Asian participants, which reduces the number of participants classified as Other Race. As a result, the second variable provides a more detailed representation of the participants' race and Hispanic origin and may therefore be more informative for describing the distribution of the study population.

```
# Age variable
age_plot <- ggplot(nhanes_data, aes(x = age)) +
  geom_histogram(binwidth = 5, fill = "gold") +
  labs(
    title = "Age Distribution",
    x     = "Age (years)",
    y     = "Count"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5)
  )

# Gender Variable
gender_plot <- ggplot(nhanes_data, aes(x = gender, fill = gender)) +
  geom_bar() +
  scale_fill_manual(
    values = c(
      "Male"   = "steelblue",
      "Female" = "deeppink"
    )
  ) +
  labs(
    title = "Gender Distribution",
    x     = "Gender",
    y     = "Count"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    legend.position = "none",
    panel.grid = element_blank(),
    axis.line = element_line(color = "grey75")
  )

# Ethnicity Variables
# Define the same colors for both ethnicity variables
ethnicity_colors <- c(
  "Mexican" = "seagreen",
  "Other Hispanic" = "yellowgreen",
  "White" = "darkolivegreen1",
  "Black" = "forestgreen",
  "Asian" = "darkgreen",
  "Other" = "olivedrab"
)
```

```

# Ethnicity 1
ethnicity1_plot <- ggplot(nhanes_data,
  aes(x = ethnicity_1, fill = ethnicity_1)
) +
  geom_bar() +
  scale_fill_manual(values = ethnicity_colors) +
  labs(
    title = "Ethnicity and Hispanic Origin Distribution (RIDRETH1)",
    x      = "Ethnicity",
    y      = "Count"
  ) +
  theme_minimal() +
  theme(
    plot.title       = element_text(hjust = 0.5, size = 8),
    legend.position  = "none",
    panel.grid.major.x = element_blank(),
    panel.grid.minor.x = element_blank(),
    panel.grid.minor.y = element_blank()
  )

# Ethnicity 2
ethnicity2_plot <- ggplot(nhanes_data,
  aes(x = ethnicity_2, fill = ethnicity_2)
) +
  geom_bar() +
  scale_fill_manual(values = ethnicity_colors) +
  labs(
    title = "Ethnicity and Hispanic Origin Distribution (RIDRETH3)",
    x      = "Ethnicity",
    y      = "Count"
  ) +
  theme_minimal() +
  theme(
    plot.title       = element_text(hjust = 0.5, size = 8),
    legend.position  = "none",
    panel.grid.major.x = element_blank(),
    panel.grid.minor.x = element_blank(),
    panel.grid.minor.y = element_blank()
  )

# Combining plots
demographic_distribution <- plot_grid(
  age_plot,
  gender_plot,
  ethnicity1_plot,
  ethnicity2_plot,
  ncol = 2
)

# Adding an overall title
demographic_distribution_final <- ggdraw() +
  draw_plot(demographic_distribution, y = 0, height = 0.90) +
  draw_label(
    "Demographic Distribution of NHANES Participants for 2013-2018",

```

```

x = 0.5,
y = 0.97,
hjust = 0.5,
vjust = 1,
fontface = "bold"
)

demographic_distribution_final

```

Demographic Distribution of NHANES Participants for 2013–2018



```

# Removing the ethnicity 1 variable
nhanes_data <- nhanes_data %>%
  select(-ethnicity_1)

```

3. Improving a Ggplot Figure

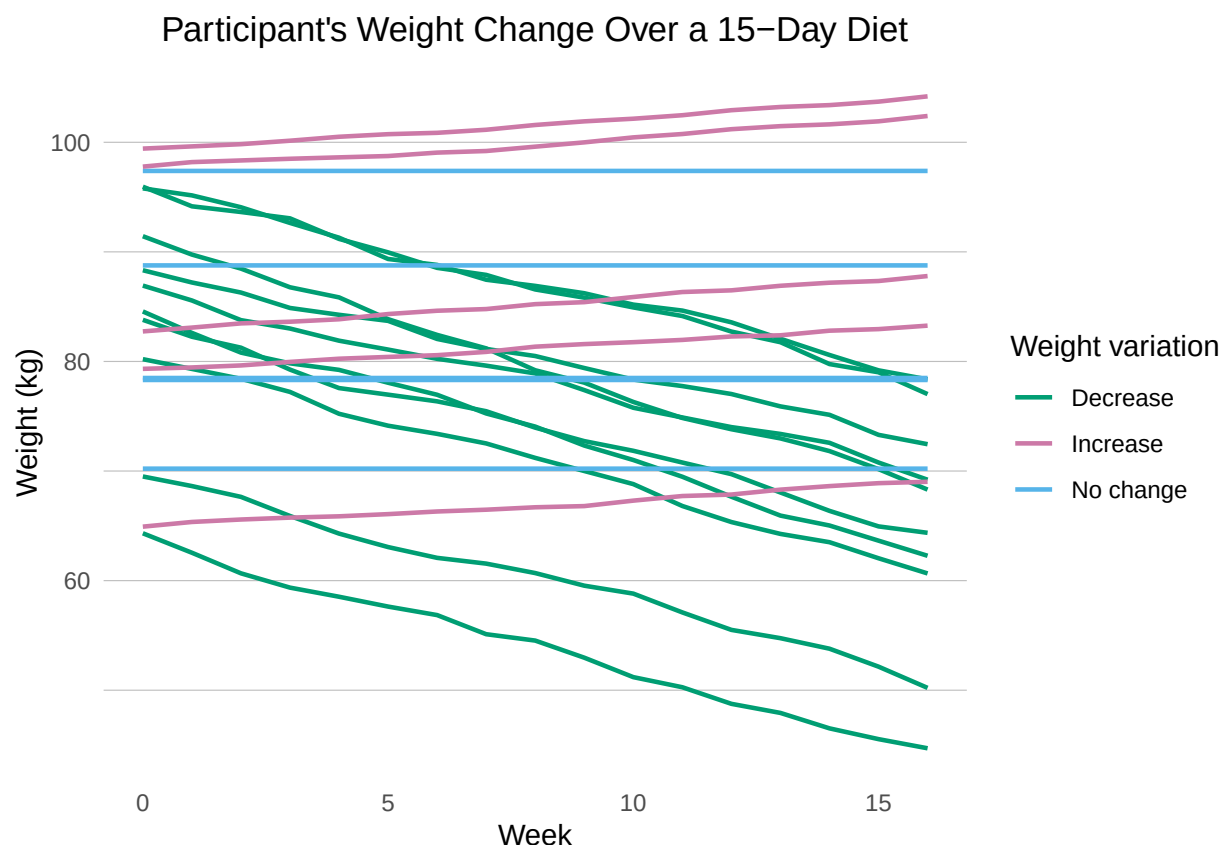
The original figure has several points that could be improved. First, the use of a rainbow color palette makes it difficult to distinguish between some individuals, such as participants 7 and 8 or 18 and 19. To make the effect of the diet on each participant's weight more apparent, the colors could instead be used to distinguish between participants who lost, gained, or maintained their weight, rather than assigning a different color to each individual. I would accomplish this by creating a new variable in the dataset that groups each participant into one of these three categories. This would make the weight changes easier to interpret while reducing the number of colors in the figure. The background also makes the figure feel somewhat crowded and does not add much information. I would therefore use simple horizontal gridlines to make changes in

patients' weight easier to follow. Finally, since weight could be reported in either kilograms or pounds, I would include the unit of measurement in the axis title to avoid confusion and add a figure title for a cleaner result.

```
# Separating the patients in three categories according to their weight variation
weight_change <- diet_data %>%
  group_by(Participant) %>%
  summarise(
    initial_weight = first(Weight),
    final_weight = last(Weight)
  ) %>%
  mutate(
    weight_variation = case_when(
      final_weight < initial_weight ~ "Decrease",
      final_weight > initial_weight ~ "Increase",
      TRUE ~ "No change"
    )
  )

# Adding the weight variation variable to the original data
diet_data <- diet_data %>%
  left_join(
    weight_change %>% select(Participant, weight_variation),
    by = "Participant"
  )

# Creating the improved plot
ggplot(diet_data, aes(x = Week, y = Weight, group = Participant, color = weight_variation)) +
  geom_line(linewidth = 0.8) +
  scale_color_manual(
    values = c(
      "Increase" = "#CC79A7",
      "No change" = "#56B4E9",
      "Decrease" = "#009E73"
    )
  ) +
  labs(
    title = "Participant's Weight Change Over a 15-Day Diet",
    x = "Week",
    y = "Weight (kg)",
    color = "Weight variation"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    panel.grid.major.x = element_blank(),
    panel.grid.minor.x = element_blank(),
    panel.grid.major.y = element_line(color = "grey", linewidth = 0.2),
    panel.grid.minor.y = element_line(color = "grey", linewidth = 0.2)
  )
```



4. Exploring Relationships through Visualizations

Introduction

The National Health and Nutrition Examination Survey (NHANES), conducted by the U.S. Centers for Disease Control and Prevention's (CDC's) National Center for Health Statistics (NCHS), is a national survey that collects data on the health of American adults and children (National Center for Health Statistics [NCHS], 2024). Between 2013 and 2018, the survey collected health data from a diverse demographic, including measurements of blood pressure (NCHS, 2024). These data allow for the examination of patterns in systolic blood pressure (SBP), including its increase with age.

Demographic and SBP Characteristics Show Expected Distributions

A first review of the distribution of participants' age, gender, average SBP, and SBP category was conducted to obtain a general overview of the sample. An average SBP variable was calculated from the available SBP measurements, and participants were then classified into SBP categories based on their average SBP. The cleaned NHANES dataset was subsequently screened for missing values in these variables. Participants with missing observations for SBP, age, or gender were removed, reducing the sample size from 29,400 to 21,604 participants. This reduction appears to have primarily affected participants under 20 years of age, although this remains the largest age group, as shown in Figure 1. The rest of the age distribution appears relatively similar, with a fairly even distribution among participants between 20 and 60 years of age, followed by a dip around age 60 and a spike at age 80. The spike at age 80 can be explained by the NHANES dataset recording ages above 80 as 80.

```

# Creating an average SBP variable
nhanes_data$mean_systolic_bp <- rowMeans(
  nhanes_data[, c("systolic_bp_1", "systolic_bp_2", "systolic_bp_3", "systolic_bp_4")],
  na.rm = TRUE
)

nhanes_data$mean_systolic_bp[
  is.nan(nhanes_data$mean_systolic_bp)
] <- NA

# Creating new dataset without missing values
nhanes_filtered <- nhanes_data %>%
  filter(
    !is.na(age),
    !is.na(gender),
    !is.na(mean_systolic_bp)
  )

# Creating new categorial variable
nhanes_final <- nhanes_filtered %>%
  mutate(
    sbp_category = case_when(
      mean_systolic_bp < 120 ~ "Normal",
      mean_systolic_bp < 130 ~ "Elevated",
      mean_systolic_bp < 140 ~ "Stage 1 HT",
      mean_systolic_bp >= 140 ~ "Stage 2 HT"
    )
  )

# Saving the NHANES dataset with the added SBP variables
rio::export(
  nhanes_final,
  here::here("Data", "cleaned_nhanes_final.csv")
)

# Comparing dataset dimensions
nrow(nhanes_data)

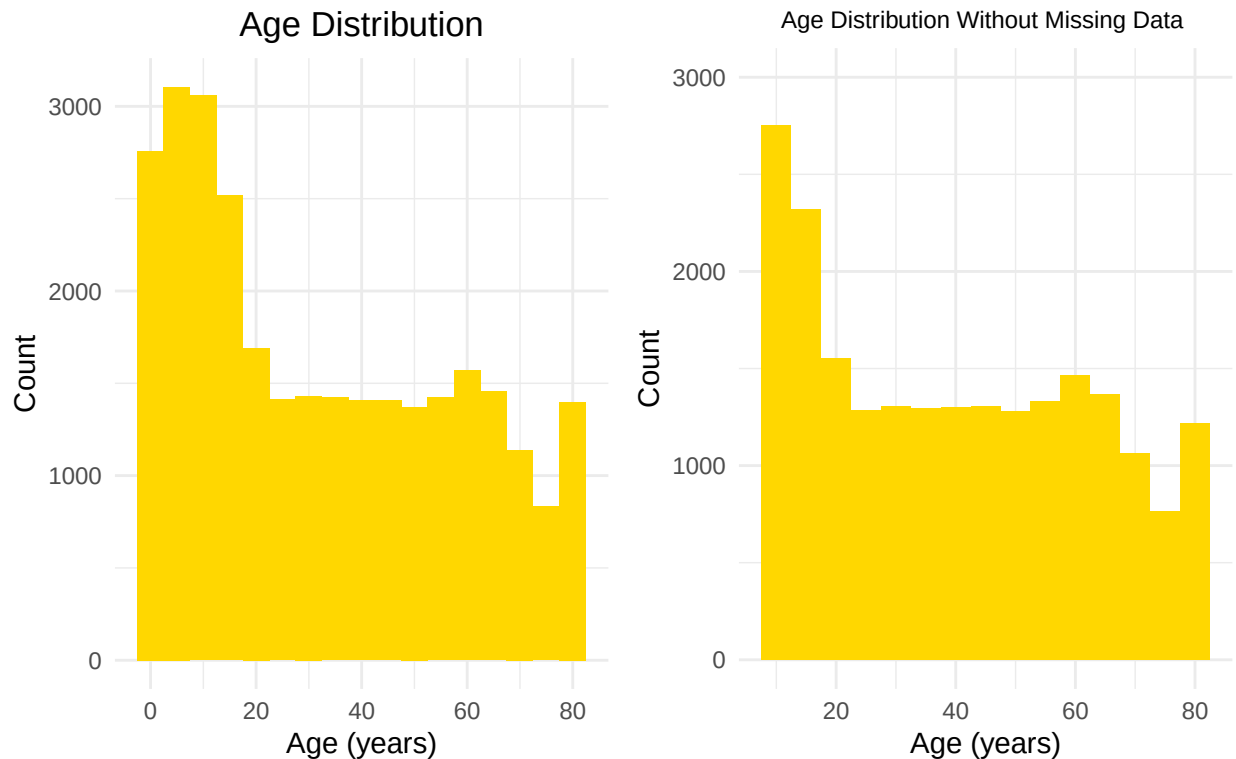
```

```
## [1] 29400
```

```
nrow(nhanes_final)
```

```
## [1] 21604
```

Figure 1. Age Distribution Before and After Screening for Missing Data



The gender distribution was relatively balanced, with slightly more female than male participants. The mean SBP was calculated from four separate readings and was then used to classify participants into the following categories : A mean SBP below 120 mmHg was classified as normal, a mean SBP from 120 to less than 130 mmHg as elevated, a mean SBP from 130 to less than 140 mmHg as Stage 1 hypertension (Stage 1 HT), and a mean SBP of 140 mmHg or higher as Stage 2 hypertension (Stage 2 HT). The majority of participants had a mean SBP between 100 and 120 mmHg, which is also reflected by the normal SBP category being the largest group.

```
# Gender distribution plot
gender_plot2 <- ggplot(nhanes_final, aes(x = gender, fill = gender)) +
  geom_bar() +
  scale_fill_manual(
    values = c(
      "Male" = "steelblue",
      "Female" = "deeppink"
    )
  ) +
  labs(
    title = "Gender Distribution",
    x = "Gender",
    y = "Count"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    legend.position = "none",
  )
```

```

    panel.grid = element_blank(),
    axis.line = element_line(color = "grey75")
  )

# SBP average
sbp_plot <- ggplot(nhanes_final, aes(x = mean_systolic_bp)) +
  geom_histogram(binwidth = 5, fill = "indianred") +
  labs(
    title = "Distribution of Mean SBP",
    x = "Mean SBP (mmHg)",
    y = "Count"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 9)
  )

# SBP categories distribution
nhanes_final$sbp_category <- factor(
  nhanes_final$sbp_category,
  levels = c(
    "Normal",
    "Elevated",
    "Stage 1 HT",
    "Stage 2 HT"
  )
)

sbp_category_plot <- ggplot(
  nhanes_final,
  aes(x = sbp_category, fill = sbp_category)
) +
  geom_bar() +
  scale_fill_brewer(
    palette = "Reds"
  ) +
  labs(
    title = "Categorization of Participants by SBP",
    x = "SBP Category",
    y = "Count"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    legend.position = "none",
    panel.grid.major.x = element_blank(),
    panel.grid.minor.x = element_blank(),
    panel.grid.minor.y = element_blank()
  )

# Combining the distributions
demographic_distribution_nhanes <- plot_grid(
  gender_plot2,
  sbp_plot,

```

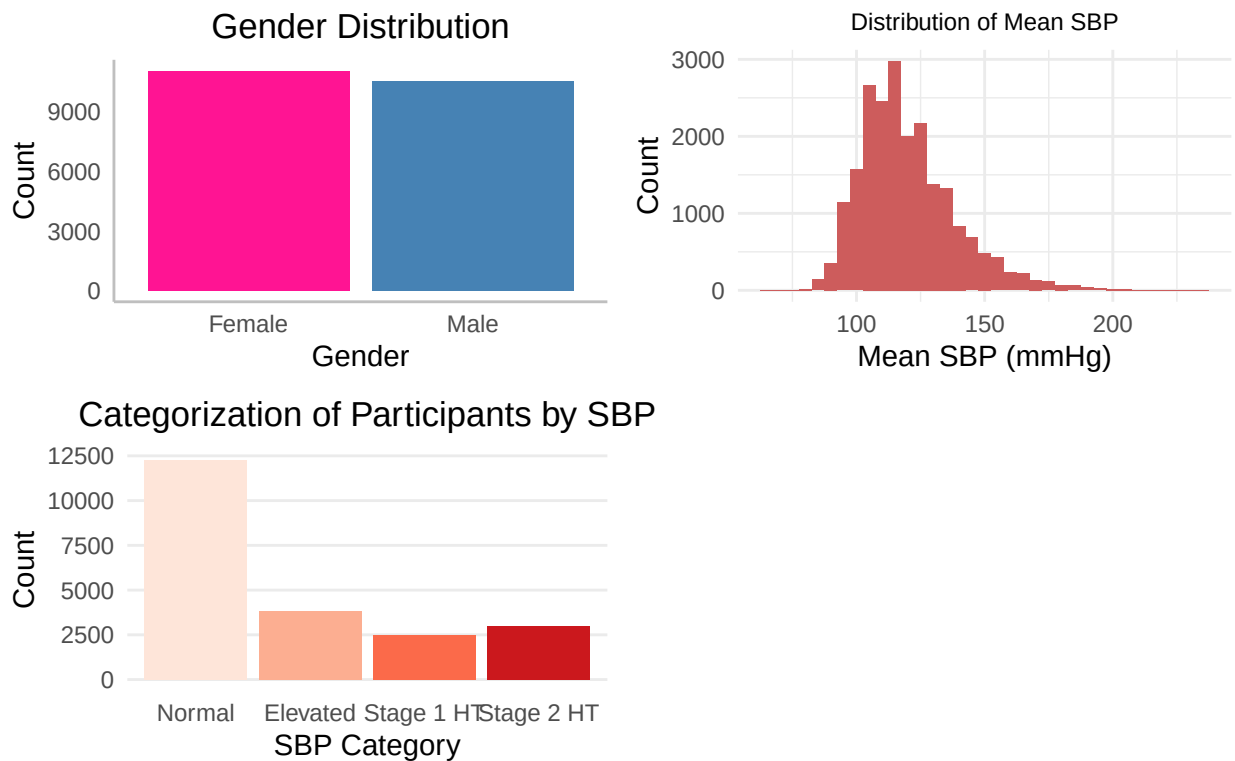
```

sbp_category_plot
)

demographic_ditribution_nhanes <- ggdraw() +
  draw_plot(demographic_ditribution_nhanes, y = 0, height = 0.90) +
  draw_label(
    "Figure 2. Demographic and SBP Characteristics of the Final Sample",
    x = 0.5,
    y = 0.97,
    hjust = 0.5,
    vjust = 1,
    fontface = "bold"
  )
)
demographic_ditribution_nhanes

```

Figure 2. Demographic and SBP Characteristics of the Final Sample



The Number of Hypertension Cases Increases With Age

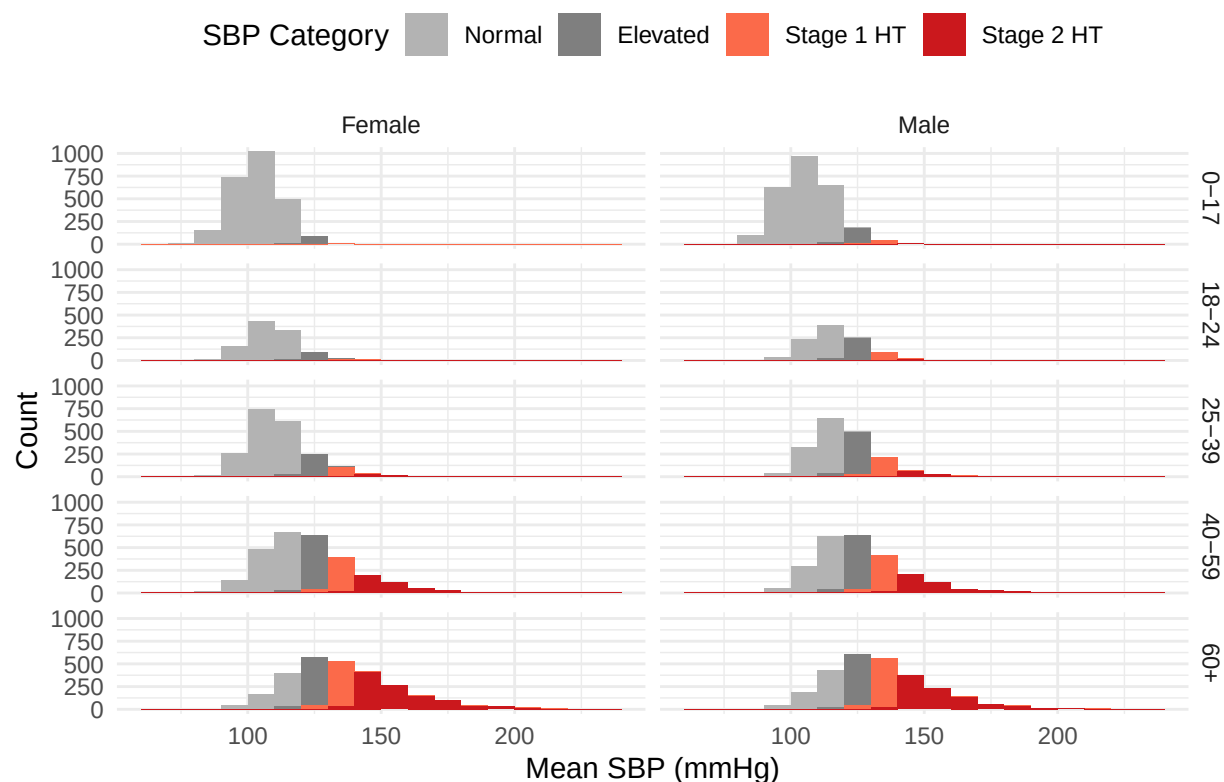
To explore the relationship between gender, age, and SBP, a faceted plot of mean SBP distribution across age groups was created. As shown in Figure 3, the distribution progressively shifts toward higher SBP values with increasing age. The figure also highlights the progressive increase in participants classified as having Stage 1 or Stage 2 hypertension. Participants aged 0–17 had almost no cases of hypertension, whereas nearly half of participants aged 60 and older were classified as having hypertension. This increase appears slightly more pronounced among males than females, suggesting a possible difference in the distribution of hypertension across genders.

```

sbp_histogram <- ggplot(
  nhanes_final,
  aes(
    x = mean_systolic_bp,
    fill = sbp_category
  )
) +
  geom_histogram(
    binwidth = 10,
    boundary = 0
  ) +
  facet_grid(
    age_cat ~ gender
  ) +
  scale_fill_manual(
    values = c(
      "Normal" = "grey70",
      "Elevated" = "grey50",
      "Stage 1 HT" = RColorBrewer::brewer.pal(4, "Reds")[3],
      "Stage 2 HT" = RColorBrewer::brewer.pal(4, "Reds")[4]
    )
  ) +
  labs(
    title = "Figure 3. Distribution of Mean SBP by Age and Gender",
    x = "Mean SBP (mmHg)",
    y = "Count",
    fill = "SBP Category"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5), legend.position = "top"
  )
sbp_histogram

```

Figure 3. Distribution of Mean SBP by Age and Gender



Conclusion

The NHANES dataset, which includes participants from a wide range of ages and a relatively balanced distribution of genders, provided SBP measurements that allowed patterns in blood pressure variation to be examined. The results showed that mean SBP and the proportion of participants classified as having hypertension increased with age. This increase appeared to be more pronounced among males than females, suggesting a possible difference in the distribution of hypertension across genders. It would be interesting to further explore whether these observed patterns in SBP and hypertension vary with ethnicity and are associated with the development of cardiovascular diseases later in life.